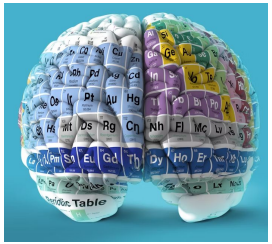
Atomic structure

Lesson Objectives

- To state mass and charge of subatomic particles.
- To describe the structure of an atom, referencing electron configuration.



Brain teaser

Thinking back to Year 8

What is the definition of an element?

Keywords: Atom, subatomic, proton, neutron, electron, electron shells, valence shell, charge, mass number, electron configuration

Atomic structure		
Meeting	Exceeding	Mastering
I can name the subatomic particles and their charges	I can explain how electrons are configured on different atoms	I can use the periodic table to draw the electron configuration for multiple atoms.

Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Atomic Structure	☐ I can name the subatomic particles and their charges	☐ I can explain how electrons are configured on different atoms	☐ I can use the periodic table to draw the electron configuration for multiple atoms.
Group 1 - The Alkali Metals	☐ I can list the group 1 elements and describe their properties as part of the alkali metals.	☐ I can explain the group 1 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the alkali metals as I read up/down the group
Group 7 - The Halogens	☐ I can list the group 7 elements and describe their properties as part of the halogens.	☐ I can explain the group 7 elements reactivity in relation to their valence shell electrons	☐ I can explain the trends in the halogens as I read up/down the group
Group 0 - The Noble Gases	☐ I can list the group 0 elements and describe their properties as part of the noble gases.	☐ I can explain the group 0 elements are inert due to their valence shell electrons	☐ I can explain the trends in the noble gases as I read up/down the group
Ionic Bonding	☐ I can describe ionic bonding as the combining of a metal atom with a nonmetal atom	☐ I can explain why metals lose electrons to because positive ions and nonmetals gain electrons to become negative ions	☐ I can use the periodic table to draw accurate dot and cross diagrams to demonstrate ionic bonds
Covalent Bonding	☐ I can describe covalent bonding as the combining of two or more nonmetal atoms	☐ I can explain how atoms will share valence electrons and form covalent molecules	☐ I can use the periodic table to draw accurate dot and cross diagrams of single, double or triple covalent bonds

Subatomic particles

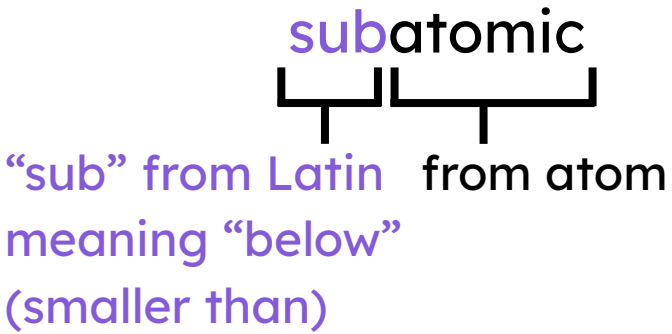
Explanation

An atom is the smallest unit of matter.

An atom is made up of three **subatomic** particles:

- **protons**
- **electrons**
- **neutrons**

Each element has a different number of subatomic particles.



Subatomic particles

Check

Name the three subatomic particles.

a atom, proton, electron

b atom, nucleus, element

c neutron, proton, electron ✓

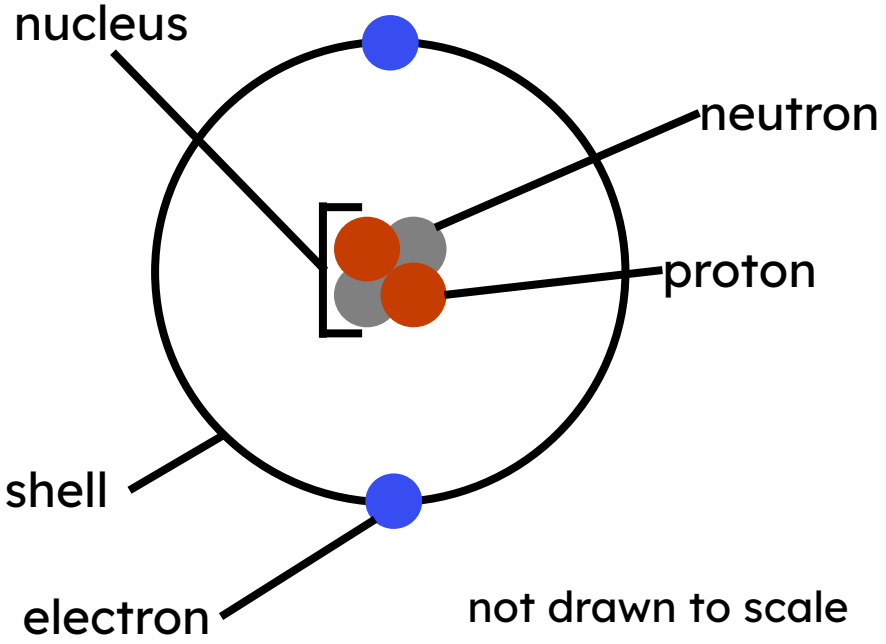
d element, electron, atom

Subatomic particles

Explanation

An atom consists of:

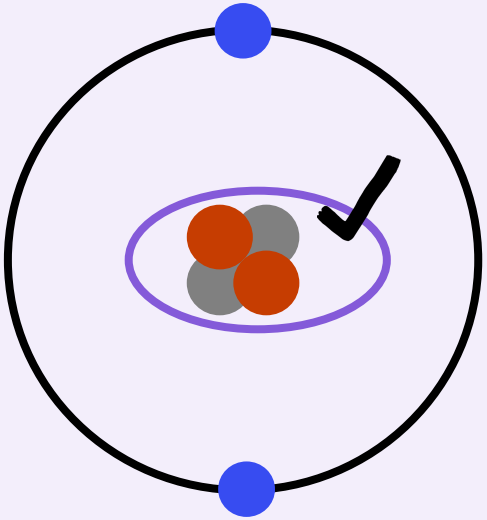
- a nucleus in the centre of the atom
- shells around the nucleus



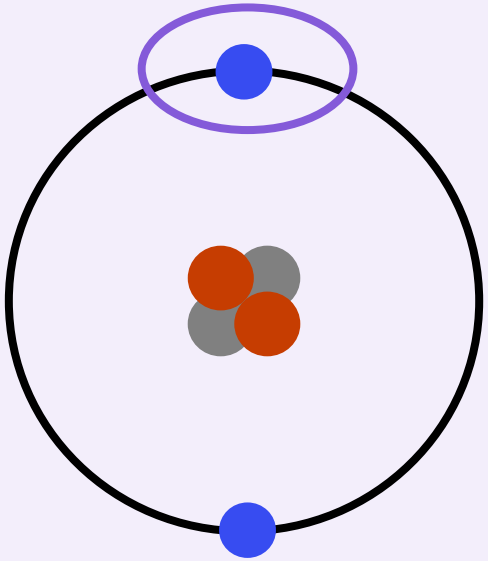
Subatomic particles

Check

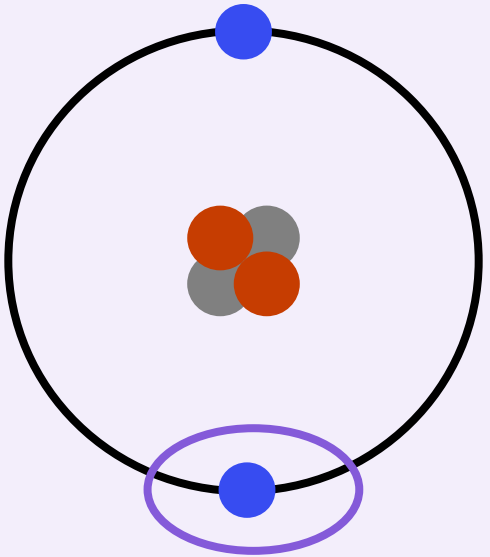
Where would you find the protons?



a



b



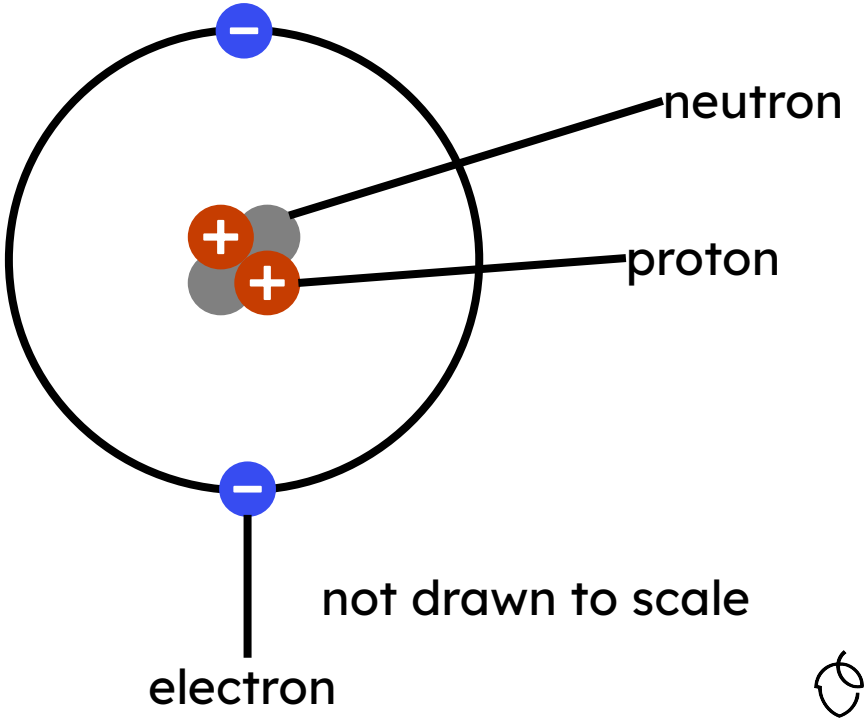
c

Subatomic particles

Explanation

The relative electric charges of subatomic particles:

- proton: +1
- electron: -1
- neutron: 0, neutral.



Subatomic particles

Check

What is the charge on an electron?

a -1 ✓

b 0

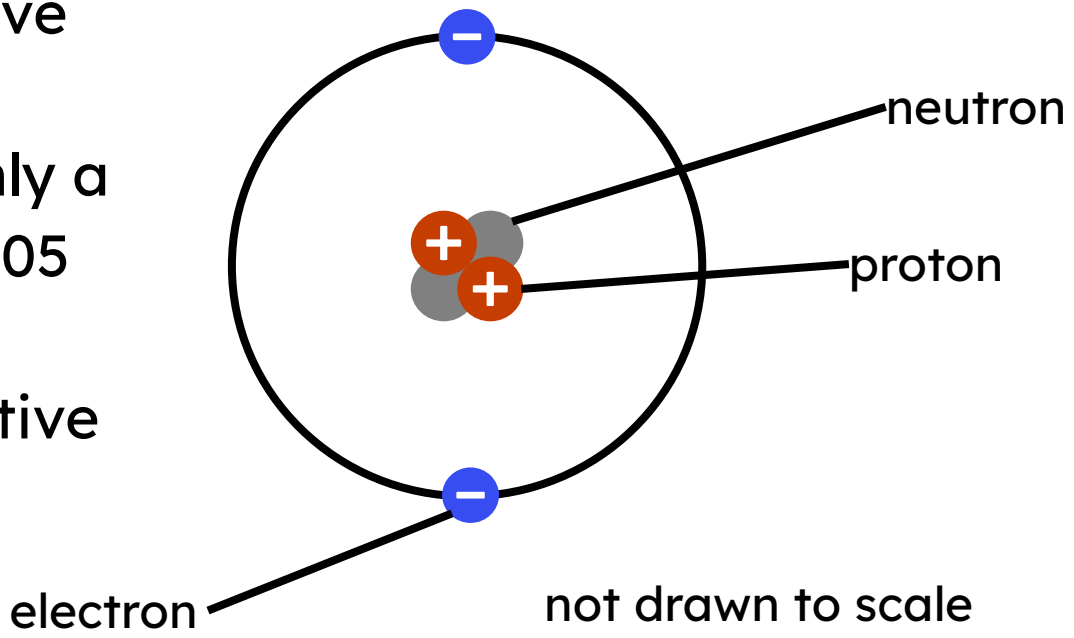
c +1

Subatomic particles

Explanation

Masses of subatomic particles:

- protons have a relative mass of 1
- electrons have roughly a relative mass of 0.0005 (very small)
- neutrons have a relative mass of 1



Subatomic particles

Check

What is the mass of an electron?

a -1

b -0.0005

c 1

d very small 0.0005 ✓

Check

True or false?

Most of the mass of an atom is found in the nucleus.

T **True** ✓ F **False**

Justify your answer

a All subatomic particles have a relative atomic mass of 1.

b The protons and neutrons are found in the centre of the atom. The electrons are found in shells. ✓

1) Copy and complete the following table.

subatomic particle	location	charge	mass
		+1	
electron			
	nucleus		1

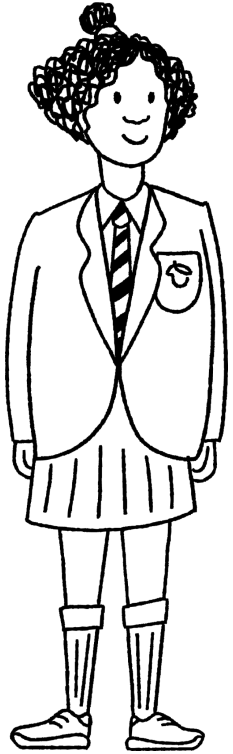
Task A Subatomic particles

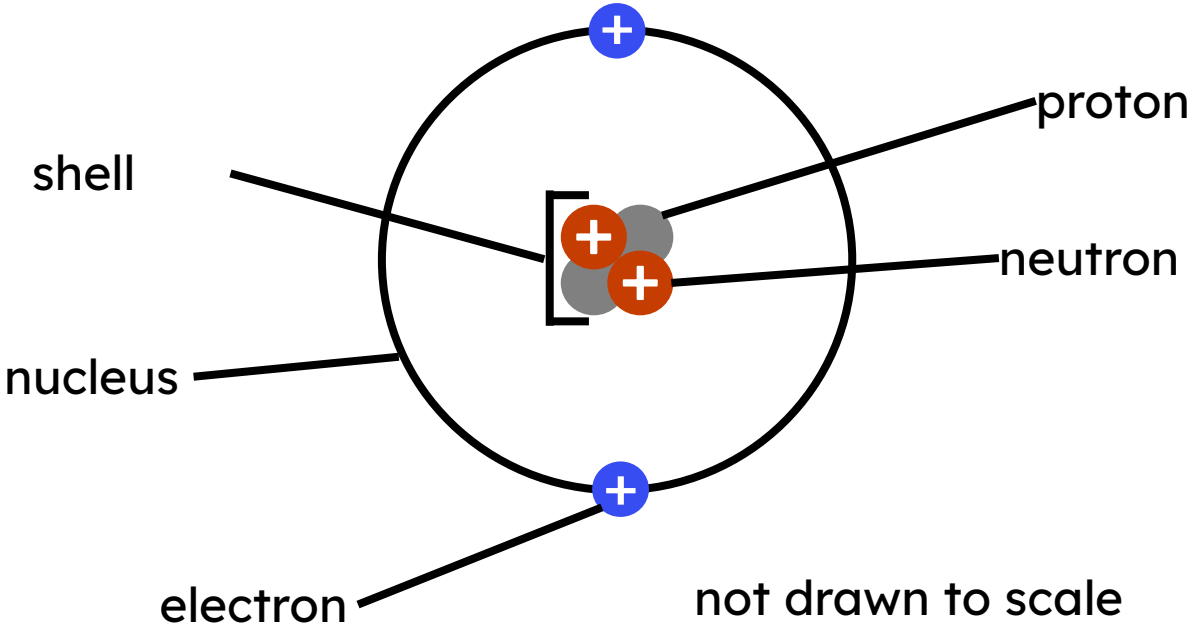
Feedback

1)

subatomic particle	location	charge	mass
proton	nucleus	+1	1
electron	shell	-1	0.0005 or very small
neutron	nucleus	0 no charge	1

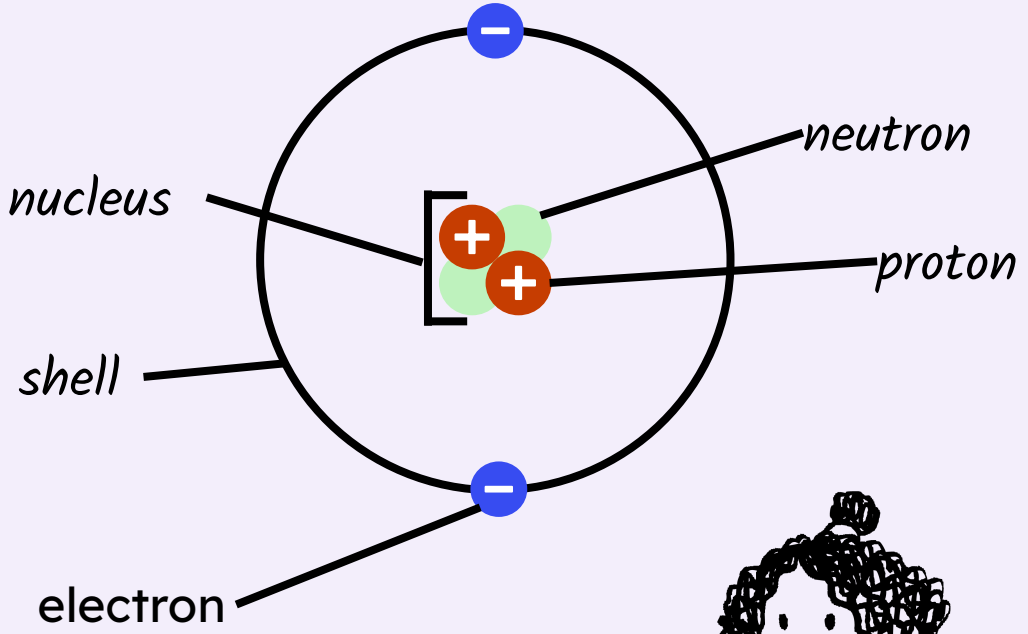
2) Correct the errors in the diagram that Izzy has drawn.





2) Errors:

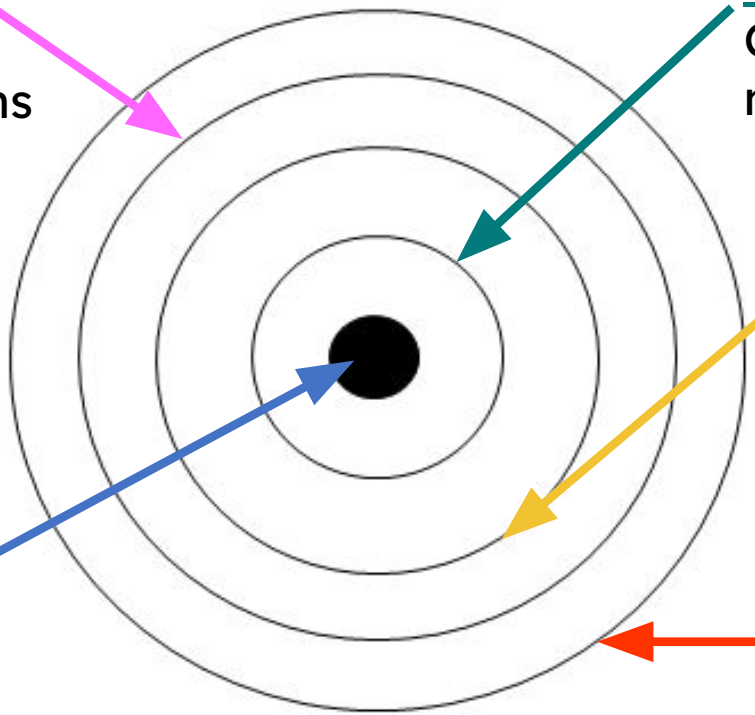
- shell and nucleus labels swapped
- proton and neutron labels swapped
- electron labelled with a + not - sign





Drawing atomic structure

Explanation



3rd Electron shell
Can contain
maximum 8 electrons

1st Electron shell
Can contain
maximum 2 electrons

2nd Electron shell
Can contain
maximum 8 electrons

Nucleus

4th Electron shell
Can contain
maximum 8 electrons

Drawing atomic structure

Explanation

3
Li
lithium
7

Step 1

Find out how many electrons does Lithium have ?

How to find it out?

Step 2

Draw a nucleus and put an electron shell around it. You can add on electron shells as we progress.

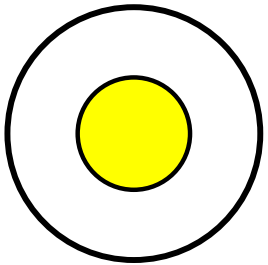
Atomic number

is same as

Proton number

is same as

Electron number

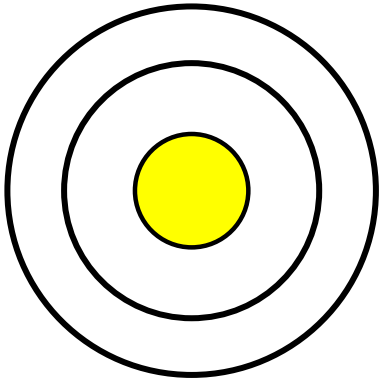


Drawing atomic structure

Explanation

Step 3

Start adding the electrons to the electron shells.



Clues:

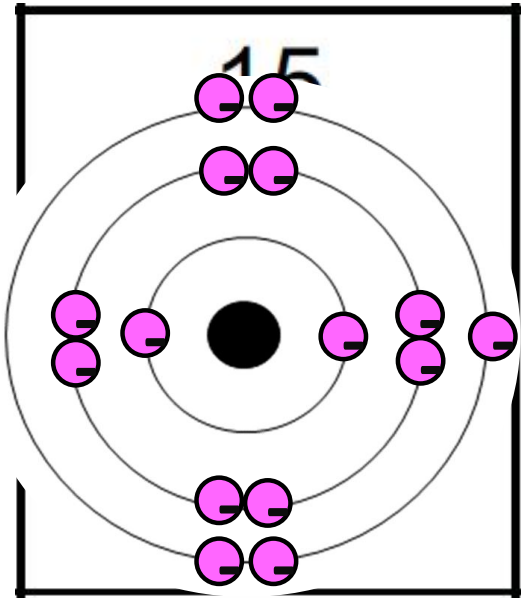
1st e- shell - max 2 electrons
2nd e- shell – max 8 electrons
3rd e- shell – max 8 electrons
4th e- shell – max 8 electrons

Total no. of electrons = 3

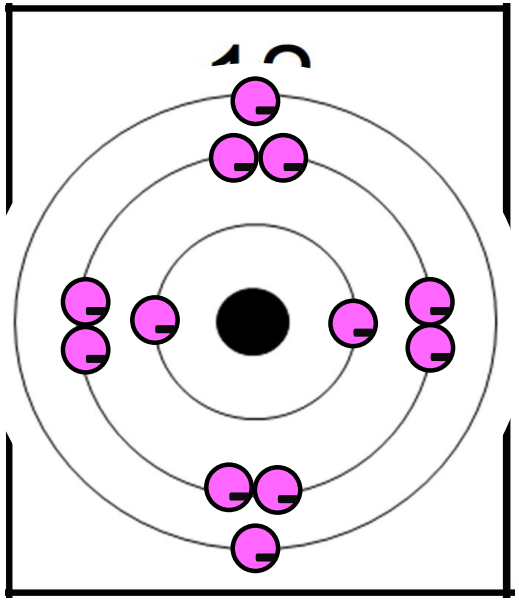
1st e- shell = 2 electrons (max)
= balance one electron

e^- Symbol for electron

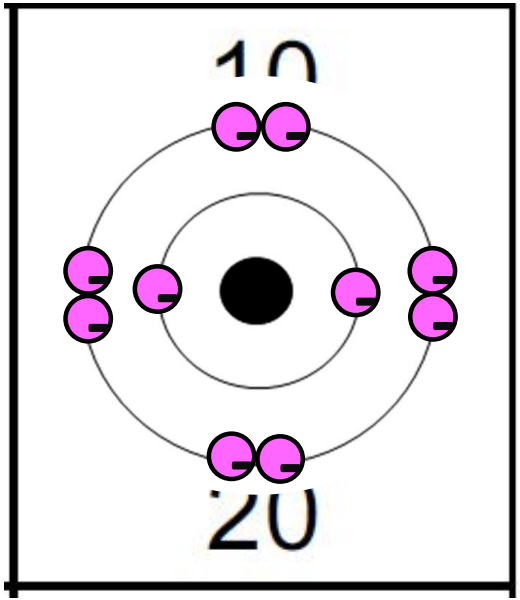
2nd e- shell = Balance electron



phosphorus



magnesium



neon

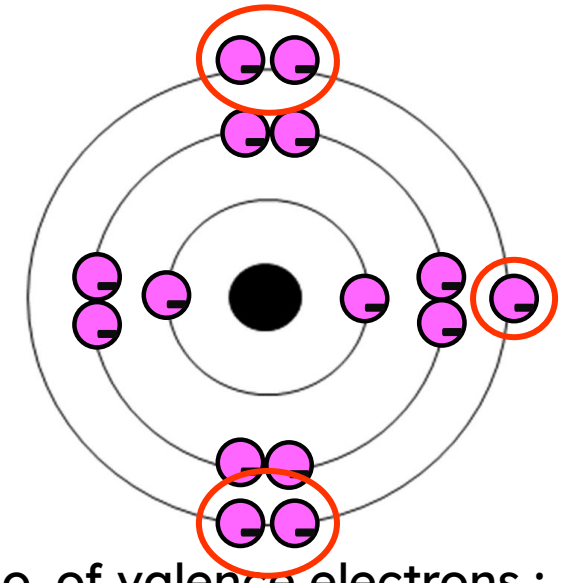
Your turn

Try to draw the atomic structure for the following elements.

Valence electrons

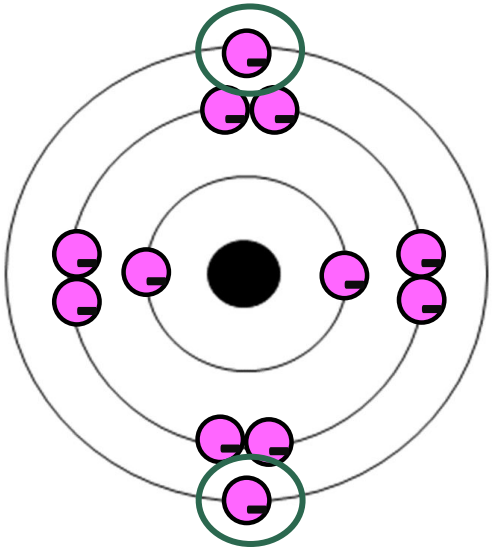
Explanation

- Valence electron refers to **the electrons in the outermost shell**.
- They are part of the atom that is **involved in a chemical reaction**.
- Determines the **chemical property** of an element.



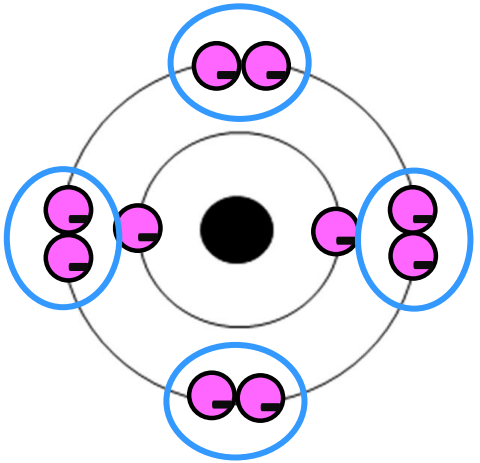
No. of valence electrons :

5



No. of valence electrons :

2



No. of valence electrons :

8

Task A The electron structure

Practice

Complete the worksheet to draw the electron structure of the the first 20 elements.



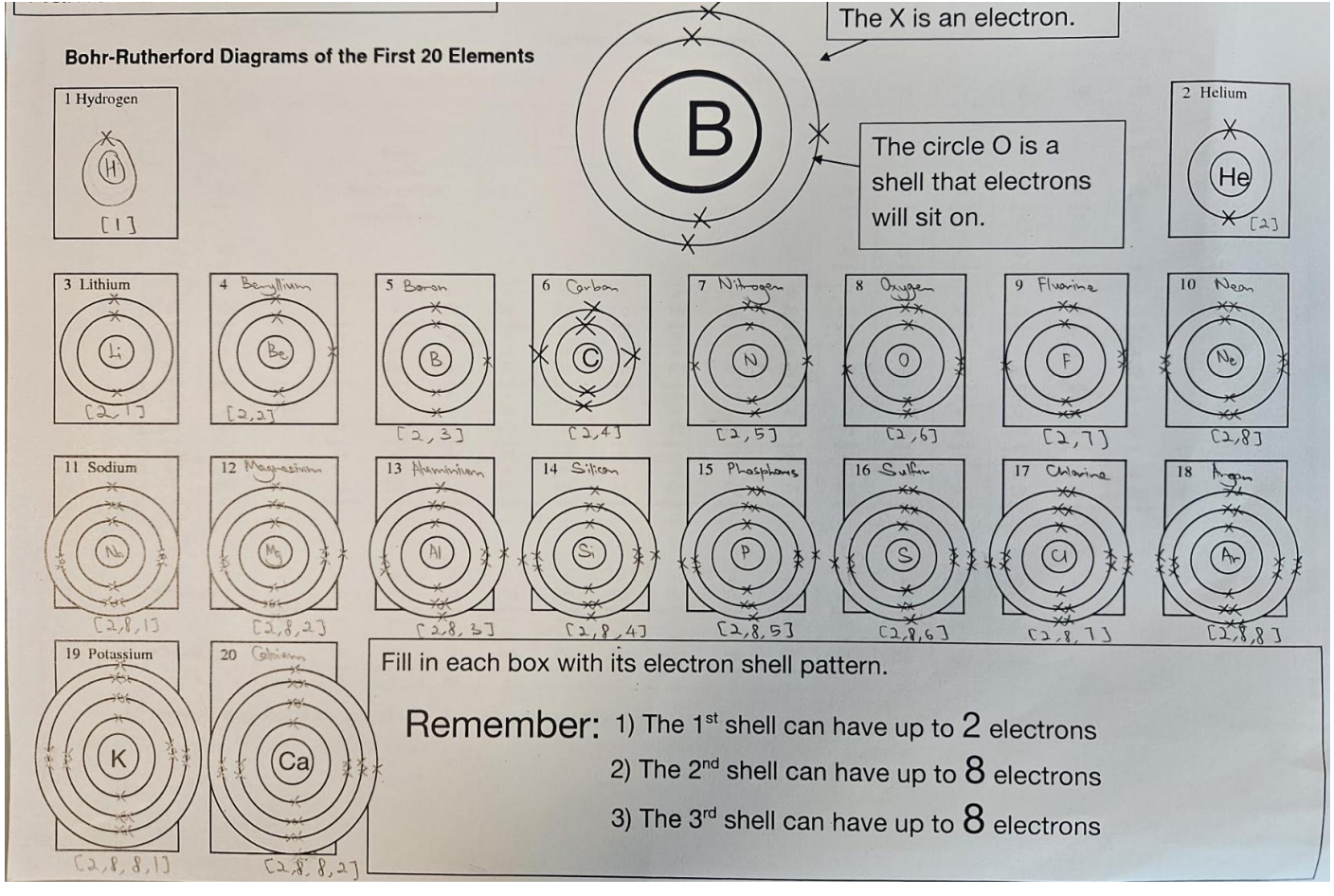
Clues:

a) First electron shell – maximum 2 electrons
b) Second electron shell – maximum 8 electrons
c) Third electron shell – maximum 8 electrons
d) Fourth electron shell – maximum 8 electrons

Hydrogen, H Structure:							Helium, He Structure:
Lithium, Li Structure:	Beryllium, Be Structure:		Boron, B Structure:	Carbon, C Structure:	Nitrogen, N Structure:	Oxygen, O Structure:	Fluorine, F Structure:	Neon, Ne Structure:
Sodium, Na Structure:	Magnesium, Mg Structure:		Aluminum, Al Structure:	Silicon, Si Structure:	Phosphorus, P Structure:	Sulfur, S Structure:	Chlorine, Cl Structure:	Argon, Ar Structure:
Potassium, K Structure:	Calcium, Ca Structure:						

Look at the completed electron structure. What link do you see between the electron structure and groups?

Task A (Answers)

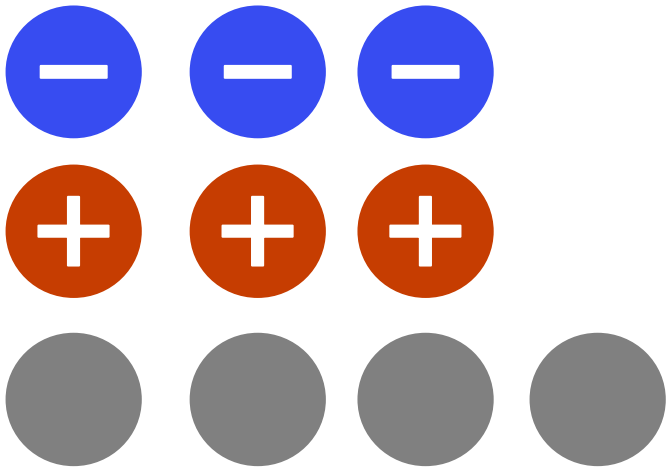


Physical characteristics of an atom

Explanation

An atom has no overall electric charge as it contains equal numbers of protons and electrons to balance the charge.

For example, a lithium atom has three electrons, three protons and four neutrons.



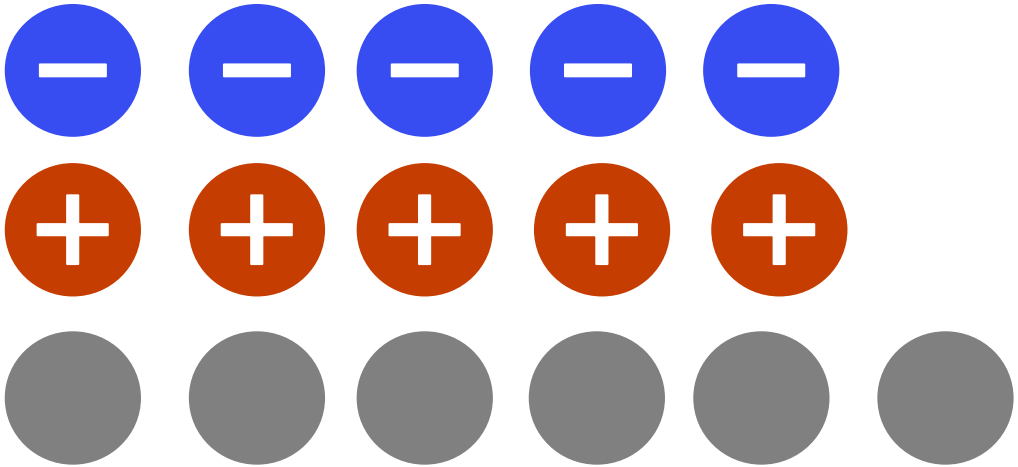
The three negative electrons cancel out the charge of the three positive protons.

As the neutrons have no charge, they do not affect the overall charge of an atom.

Physical characteristics of an atom

Explanation

A boron atom has 5 electrons, 5 protons and 6 neutrons.



The 5 negative electrons cancel out the charge of the 5 positive protons.

As the neutrons have no charge, they do not affect the overall charge of an atom.

Physical characteristics of an atom

Check

An aluminium atom has 13 electrons, 13 protons and 14 neutrons. What is the overall charge of an aluminium atom?

a -1

b 0 ✓

c +1

d +26

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